# Problem Set M

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# Bellman Equation

- The state variable here is the realized shock on demand function, let's say it's . And the control variable is the price . Then the Bellman Equation is just:

$$\Pi(u) = \max_{p}\{\pi(p) + \beta E(\Pi(u')|u)\}$$

- However in our model, some parameters in $\pi(p)$ are unknown. (we will solve stochastic demand in the 2nd part)

# Bayesian demand parameters

- We are trying to solve:

$$\Pi(P) = \max_{\{p\}} \frac{1}{1-\beta} E[p * D(p) - C(D(p))]$$

- Which is equivalent to maximizing one periods profits given our belief about the parameters of demand.
- Normally we use first order conditions, however …

# FOC by Mathematica

- The first order condition is:

$$e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1} + e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1}\,(b1 + 2\,b2\,\mathrm{Log}[p]) - \frac{2.21672\,e^{0.1\,\mathrm{Log}\left[e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1}\right]^2}\left(e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1}\right)^{0.3}(b1 + 2.\,b2\,\mathrm{Log}[p])}{p} - \frac{1.47781\,e^{0.1\,\mathrm{Log}\left[e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1}\right]^2}\left(e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1}\right)^{0.3}(b1 + 2.\,b2\,\mathrm{Log}[p])\,\mathrm{Log}\left[e^{b0+b2\,\mathrm{Log}[p]^2}\,p^{b1}\right]}{p}$$

- However it’s not solvable algebraically.

# Numerical Maximization

- We use Kwopt's example demand parameters as our prior:

| (Intercept) | $log(p)$ | $I(log(p)^2)$ |
|---|---|---|
| 4.7856373 | -0.3113628 | -0.2160528 |

- For the point estimate of maximum profit we numerically get 144.58, which is achieved at price 9.30.

# Bootstrapping 1

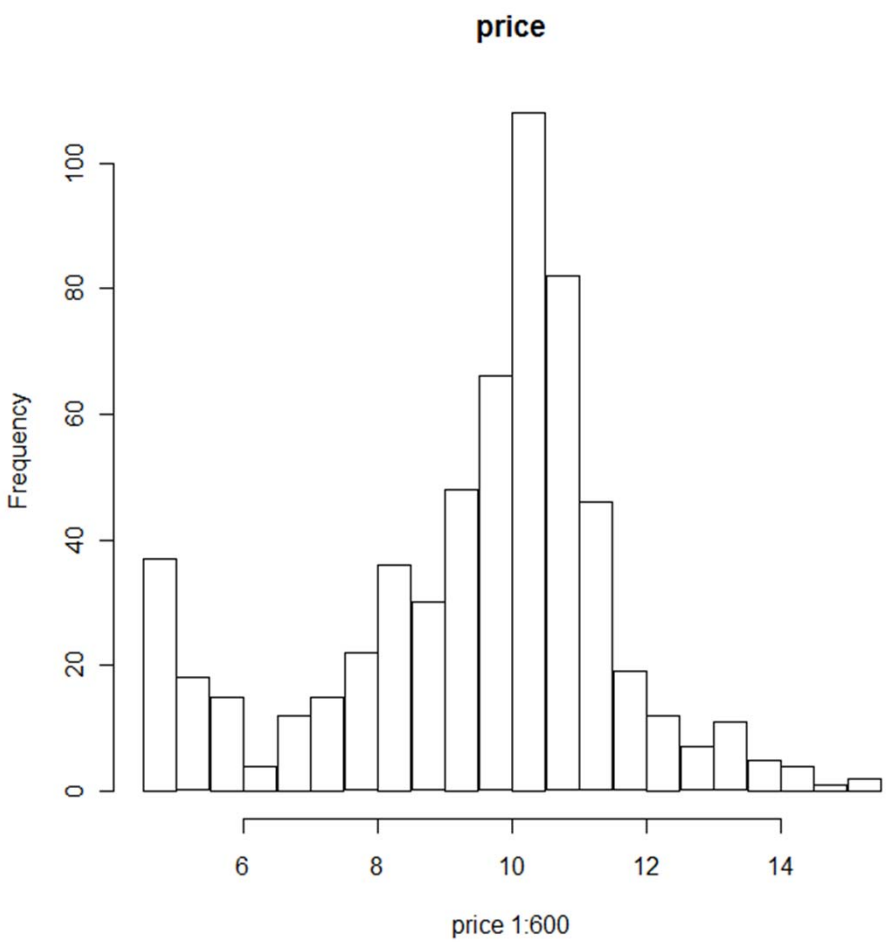
1. We draw random ordered numbers from multivariate normal distribution for $\beta_0, \beta_1, \beta_2$.
2. Each time we maximize profit using numerical methods and save the corresponding price and profit →

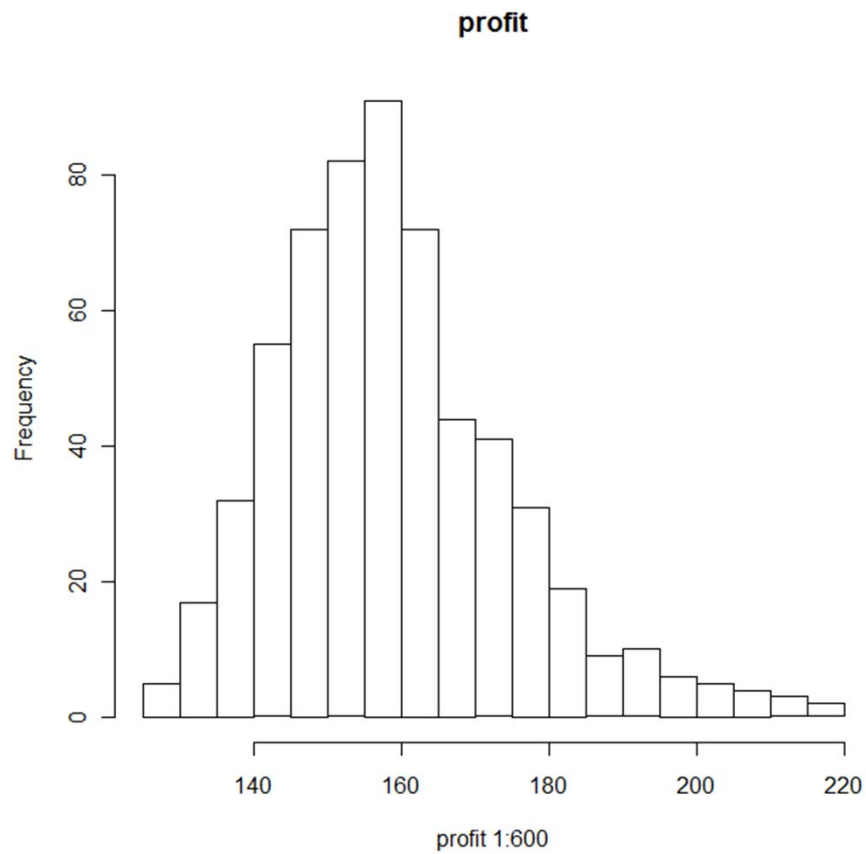
| summary | |
|---|---|
| Min. | $4.599e+00$ |
| 1st Qu. | $9.623e+00$ |
| Median | $1.111e+01$ |
| Mean | $1.935e+14$ |
| 3rd Qu. | $8.263e+01$ |
| Max. | $1.935e+17$ |

| summary | |
|---|---|
| Min. | $1.258e+02$ |
| 1st Qu. | $1.543e+02$ |
| Median | $1.740e+02$ |
| Mean | $6.234e+22$ |
| 3rd Qu. | $2.968e+05$ |
| Max. | $6.234e+25$ |

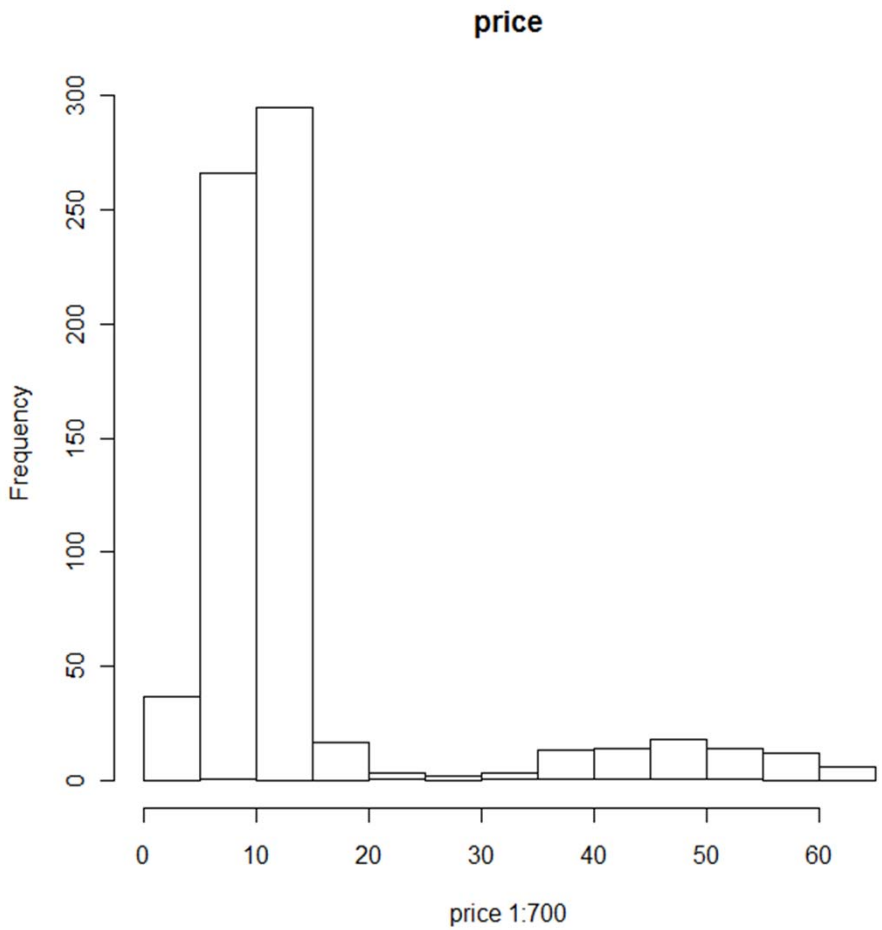
- Looking at quartiles and maximum profit → positive sloping demands!

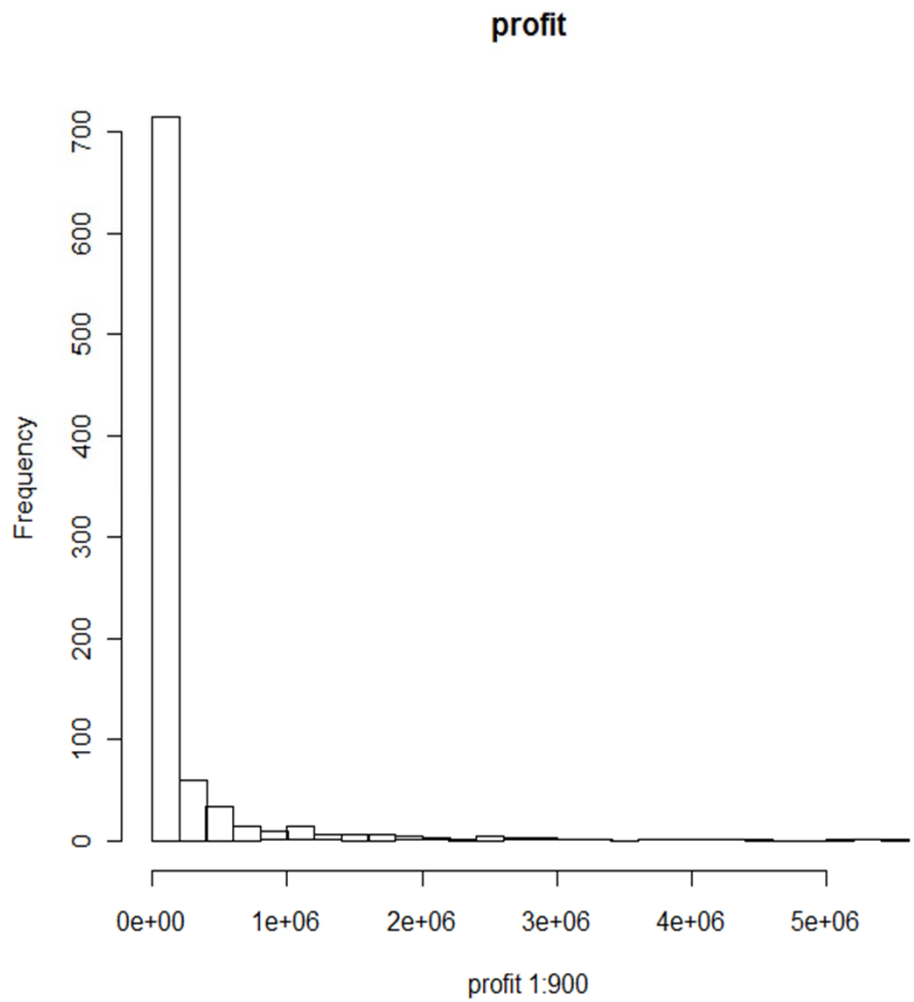
# Bootstrapping 2 → 60% of observations histogram



# Bootstrapping 3



# Infinitely Patient Firm

- $\beta = 1$ , then we now have no trade-off, because the future value would not be less valuable. So we are free to get more information about the demand function by doing price experiments for several periods.
- So in order to recover the whole demand curve, we can try so many price experiments to recover demand completely and then maximize profit.

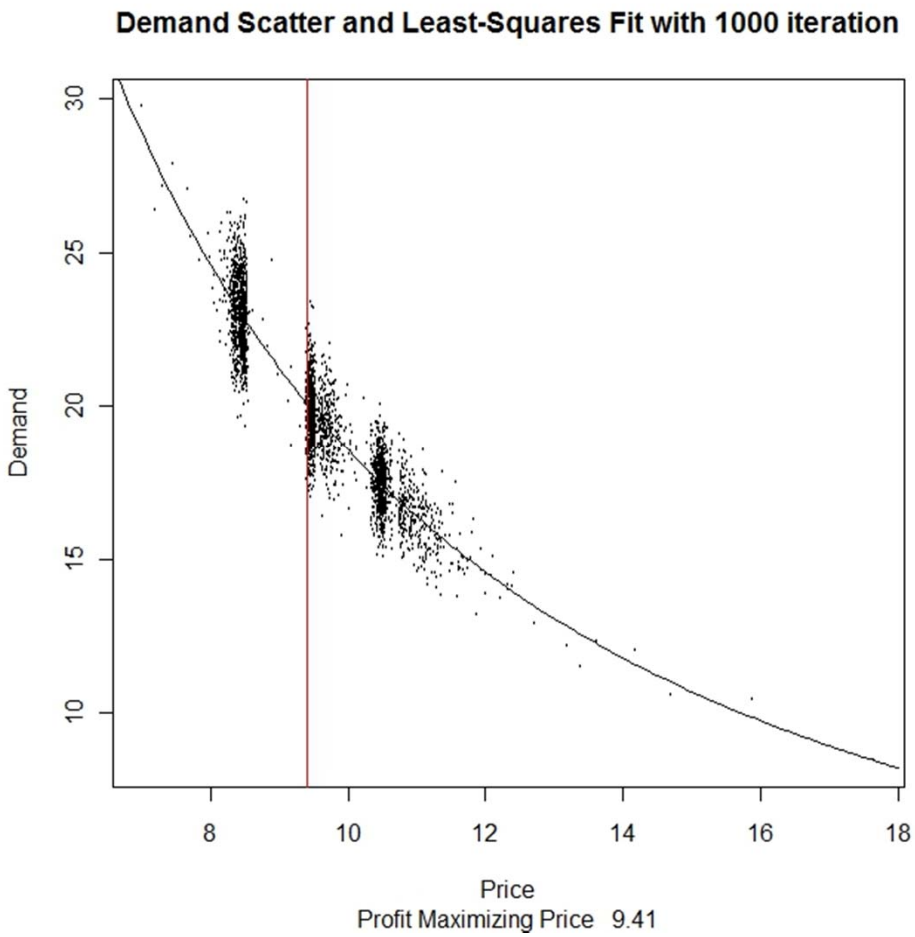
# Analyzing KW optimization

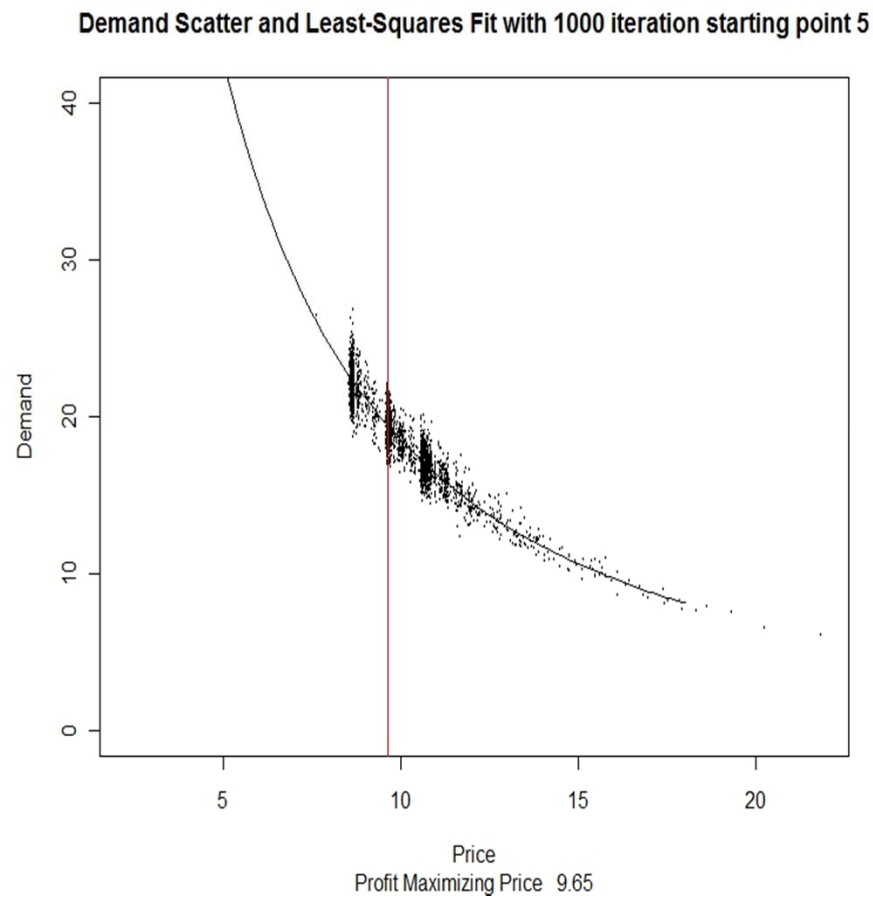
Optimal price given different initial prices after 100 KW iterations. →

| initial price | $P^*$ |
|---|---|
| 4 | 21.47802661973236 |
| 5 | 10.72961463073115 |
| 6 | 9.50178554434274 |
| 7 | 10.17277225219732 |
| 8 | 9.20022723758826 |
| 9 | 9.59144158747898 |
| 10 | 9.87695846929285 |
| 11 | 9.57903385522460 |
| 12 | 9.76744650087086 |
| 13 | 9.72447328238406 |
| 14 | 9.06873631988005 |
| 15 | 9.84253277451968 |
| 16 | 9.72792642340899 |
| 17 | 10.02088193545897 |
| 18 | 9.95745332716329 |
| 19 | 10.54956423930031 |
| 20 | 10.97167394064366 |
| 21 | 12.32754253721391 |
| 22 | 13.06549436106232 |
| 23 | 13.32526191404650 |
| 24 | 12.22976928912359 |

This table doesn't change much even with 1000 iterations.

- As for estimating demand function, we can use 300 price and profit observations after a 100-iteration Kwopt:



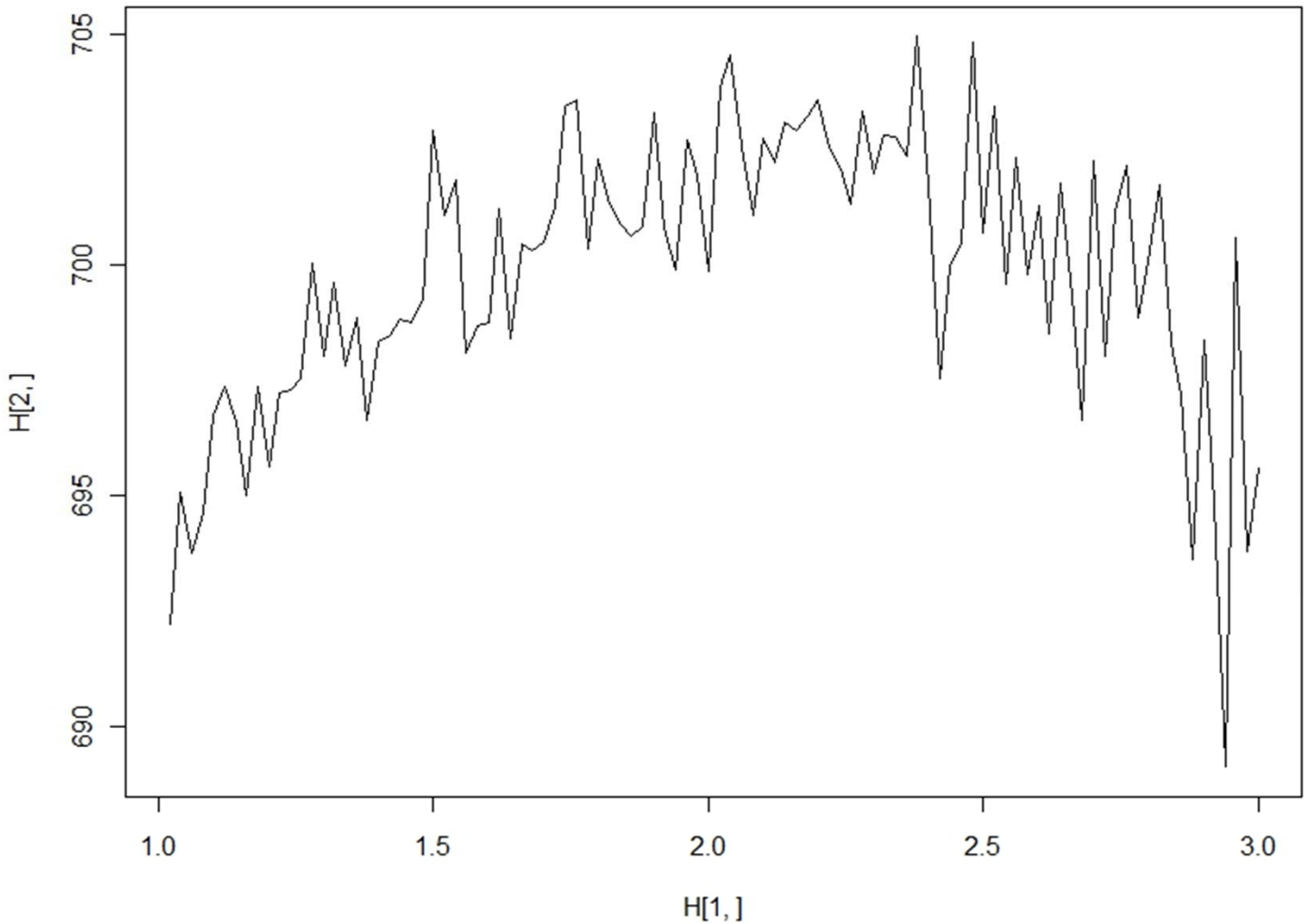



# Which KW optimization parameters?

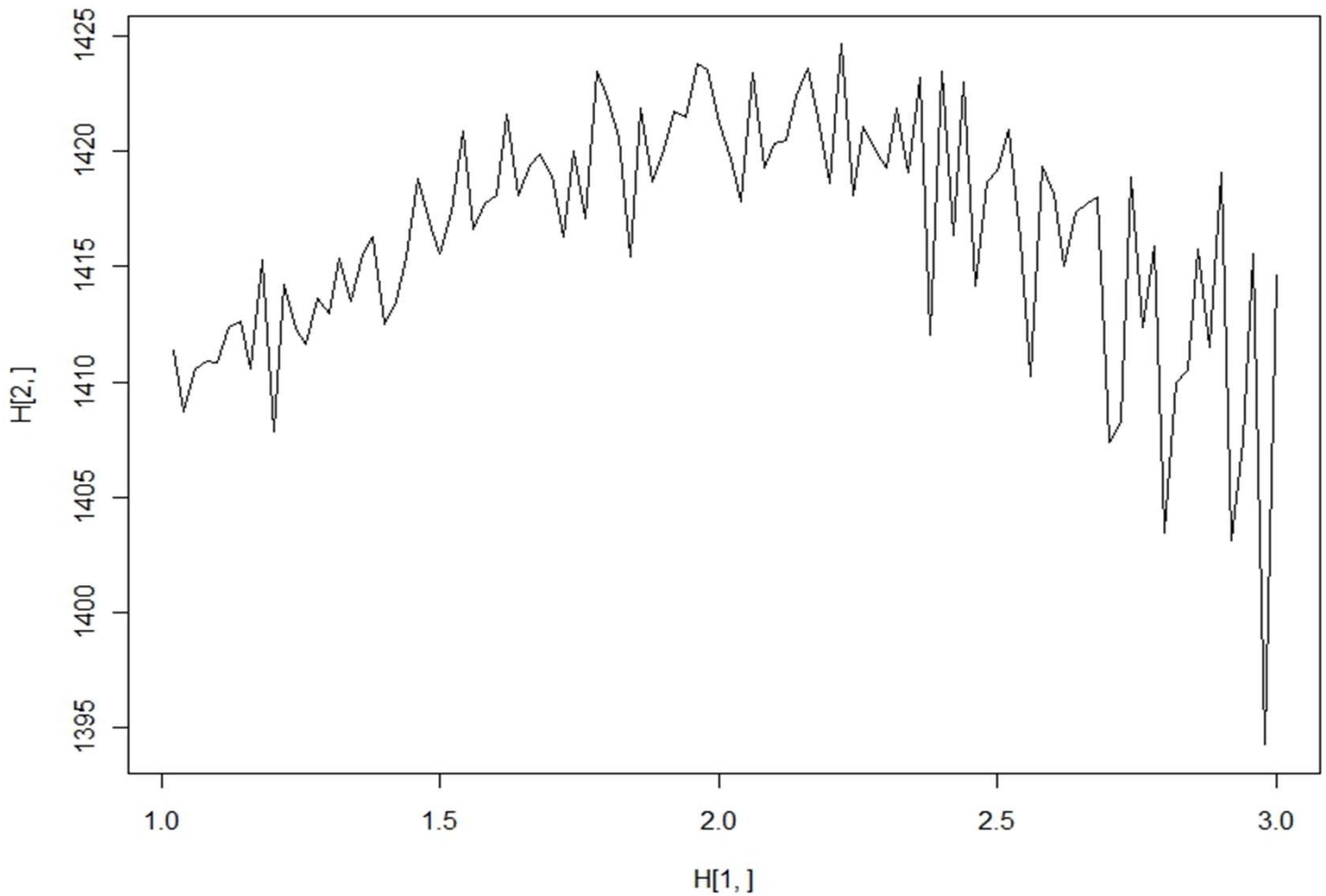
- To figure out how parameters of the KW optimization procedure affect our discounted profit we use the below procedure:

1. Calculate mean of 100 discounted profits with the initial parameter.
2. Calculate mean of 100 discounted profits with the initial parameter plus 0.01
3. Repeat this 100 times
4. Plot the different values of parameters against means of discounted profits.
5. We can also find the parameter that has maximized the discounted profit but as the results are stochastic, quantiles and plot are more informative.

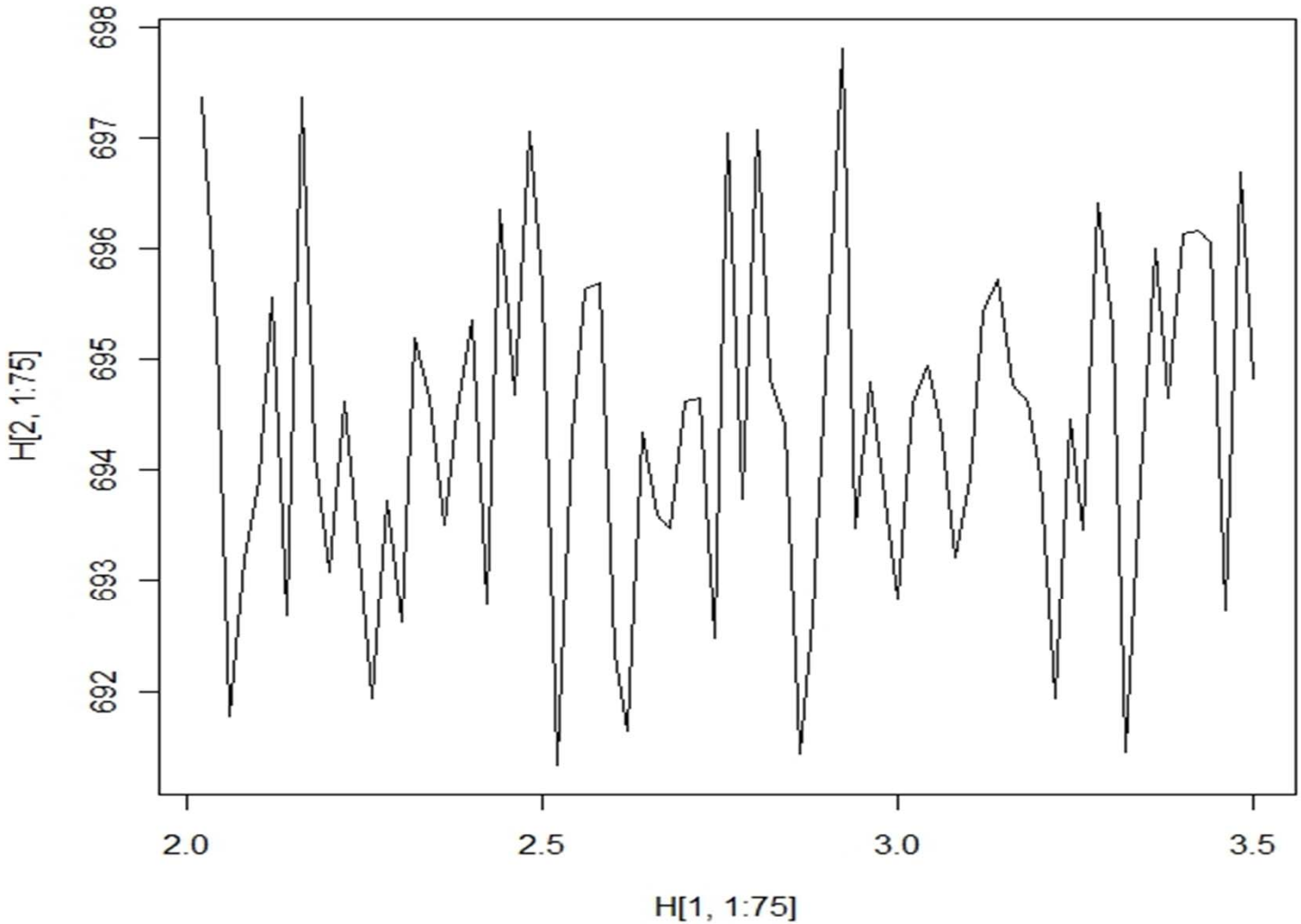
# Find best a0 with β=0.8



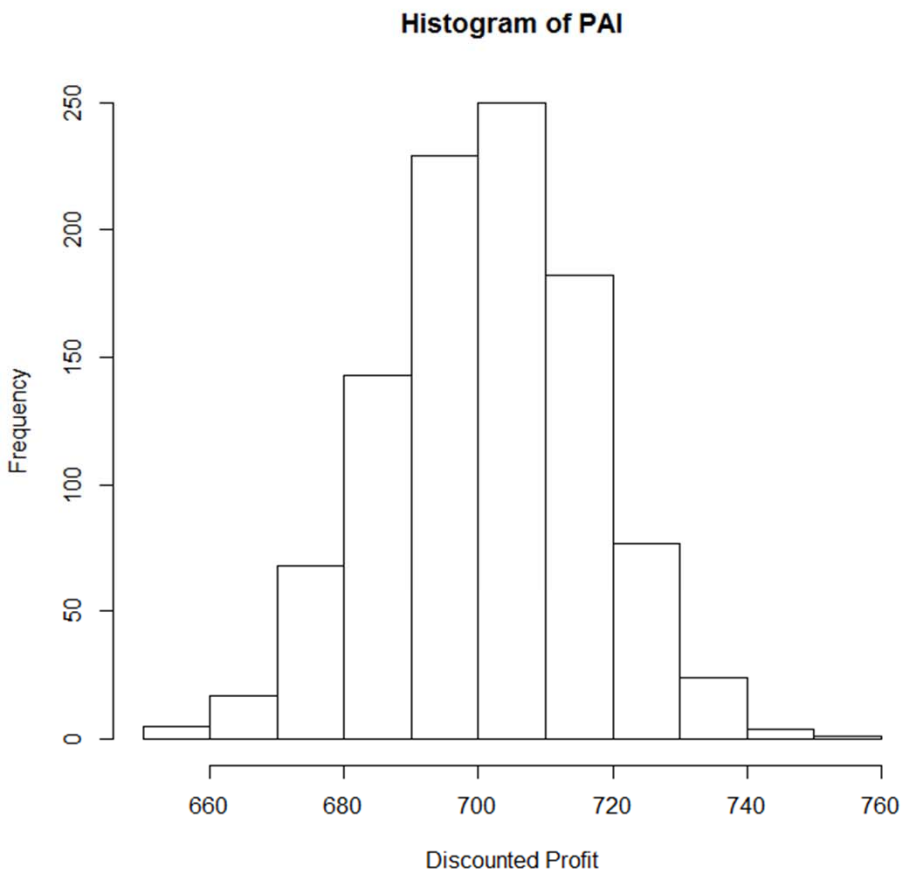
# Find best a0 with β=0.9

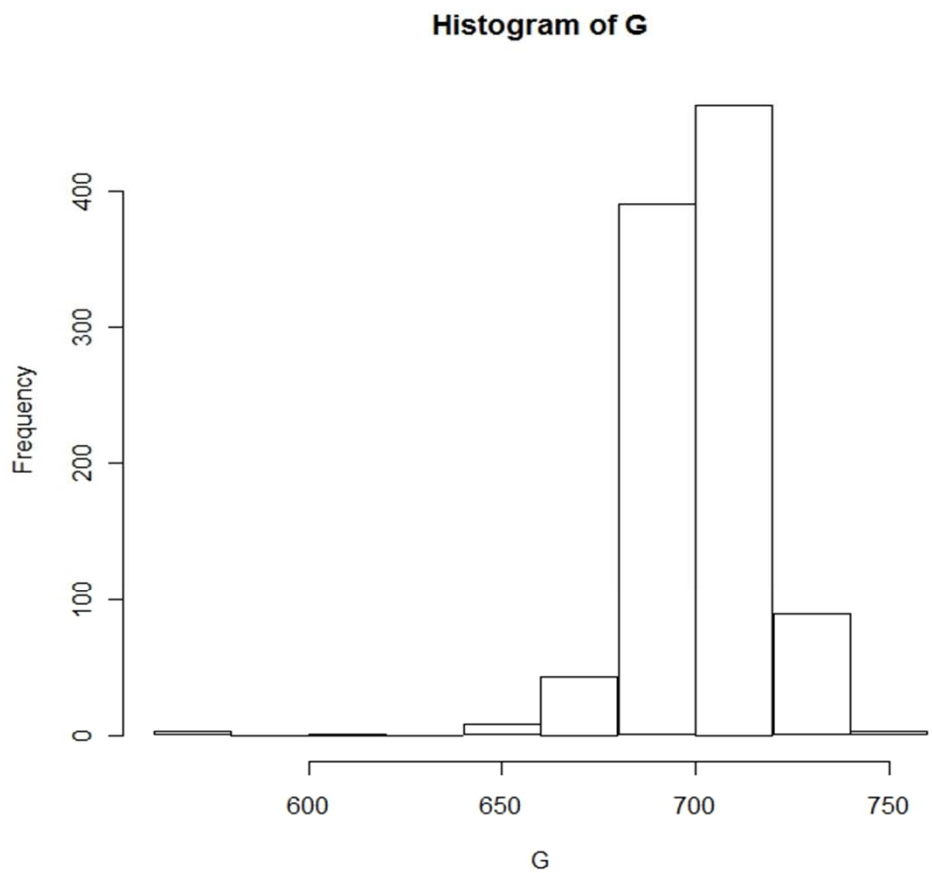


# Which c0 with β=0.8

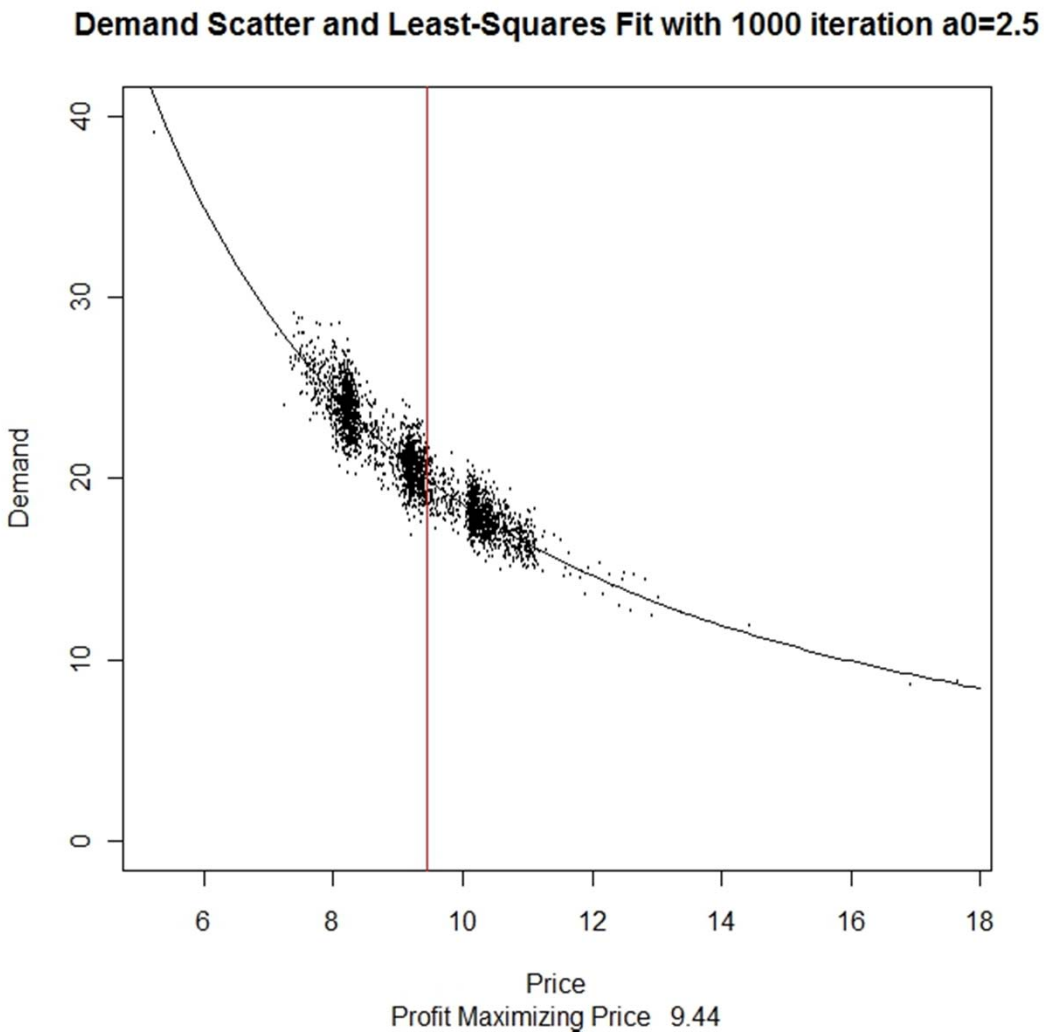


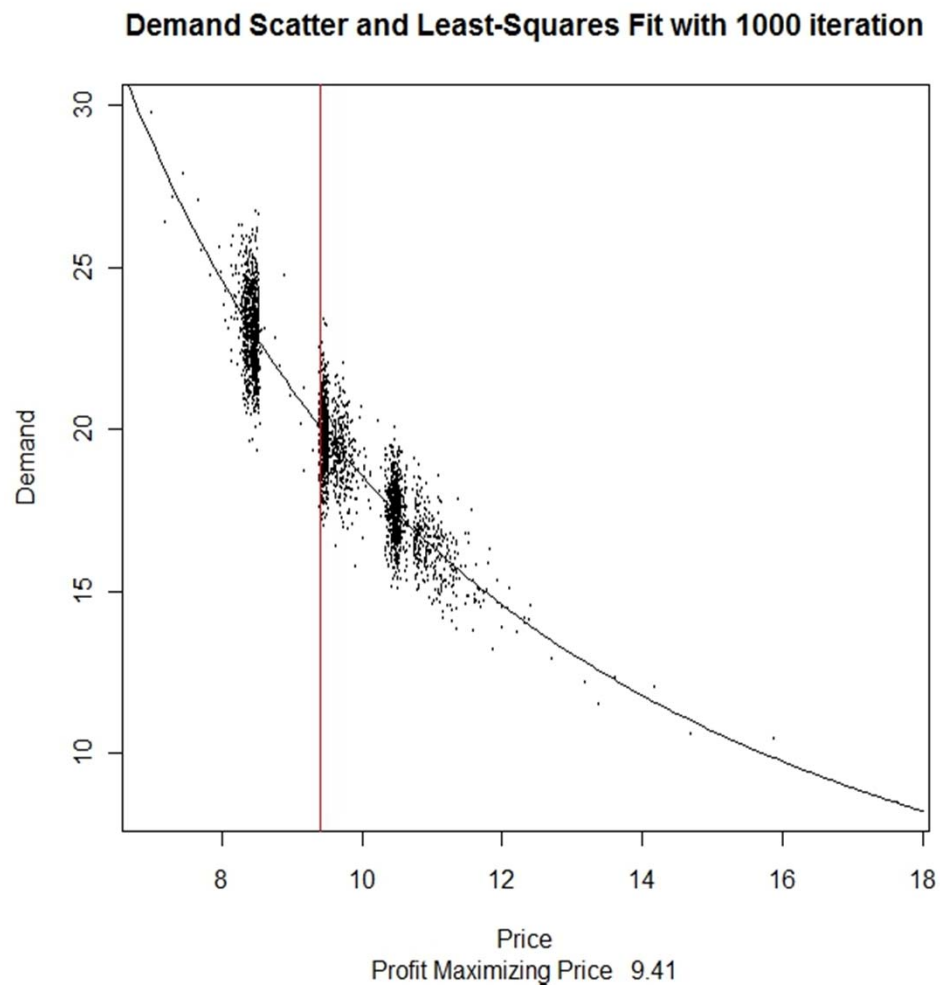
# Maximum profit (n=100, $\beta$=0.8 and P0=15) → histogram of 1000 discounted profits with a0=1 and a0=2.5(on the right panel):



# Trying a0=2.5 → move in the same direction for the two goals



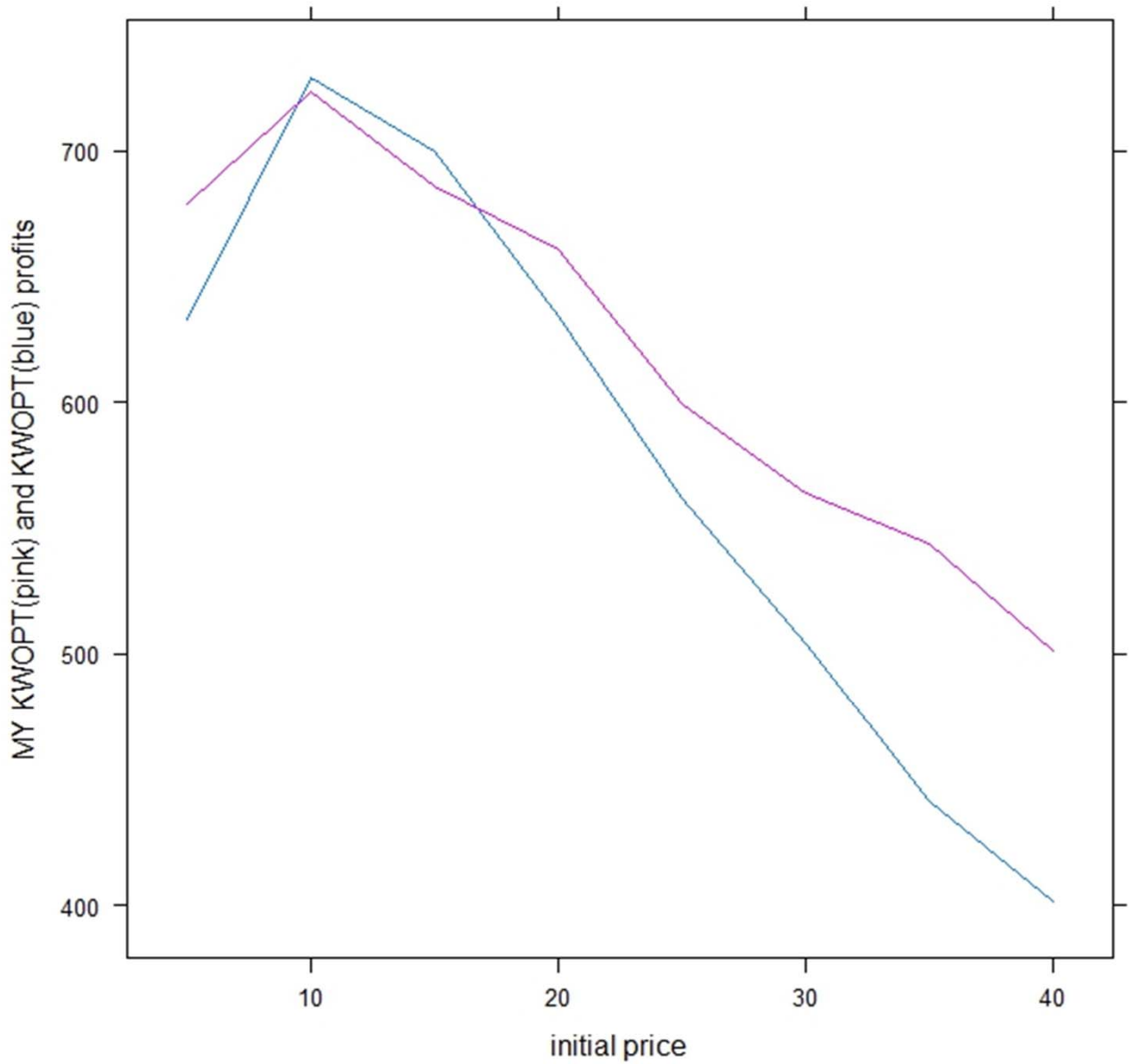
# Modifying Demand

- We've figure out that the discounted sum of profits, using the KW optimization procedure is most sensitive to the initial price.
- As a remedy, our final solution involves, using all of the data from previous periods, instead of looking at two points at a time.
- To do this we employ periodic regressions i.e. 2 times during the first $6^{th}$ periods in the middle of KW optimization process and then continue with KW optimization procedure for the rest of the periods.

# Our Procedure in steps

1. In fact we have 6 observations by 3rd period of KW procedure.
2. We use this 6 observations to estimate demand, given the specified model (in real world, the monopolists can try different specifications and see which one works best, etc.)
3. Then, we employ numerical optimization of profit function given these estimates of demand parameters, using the same method in question 2.
4. Then set the P0 of the rest of the KW optimization process to the price that we have found.

# Modified KW VS. Original KW



# Appendix 1

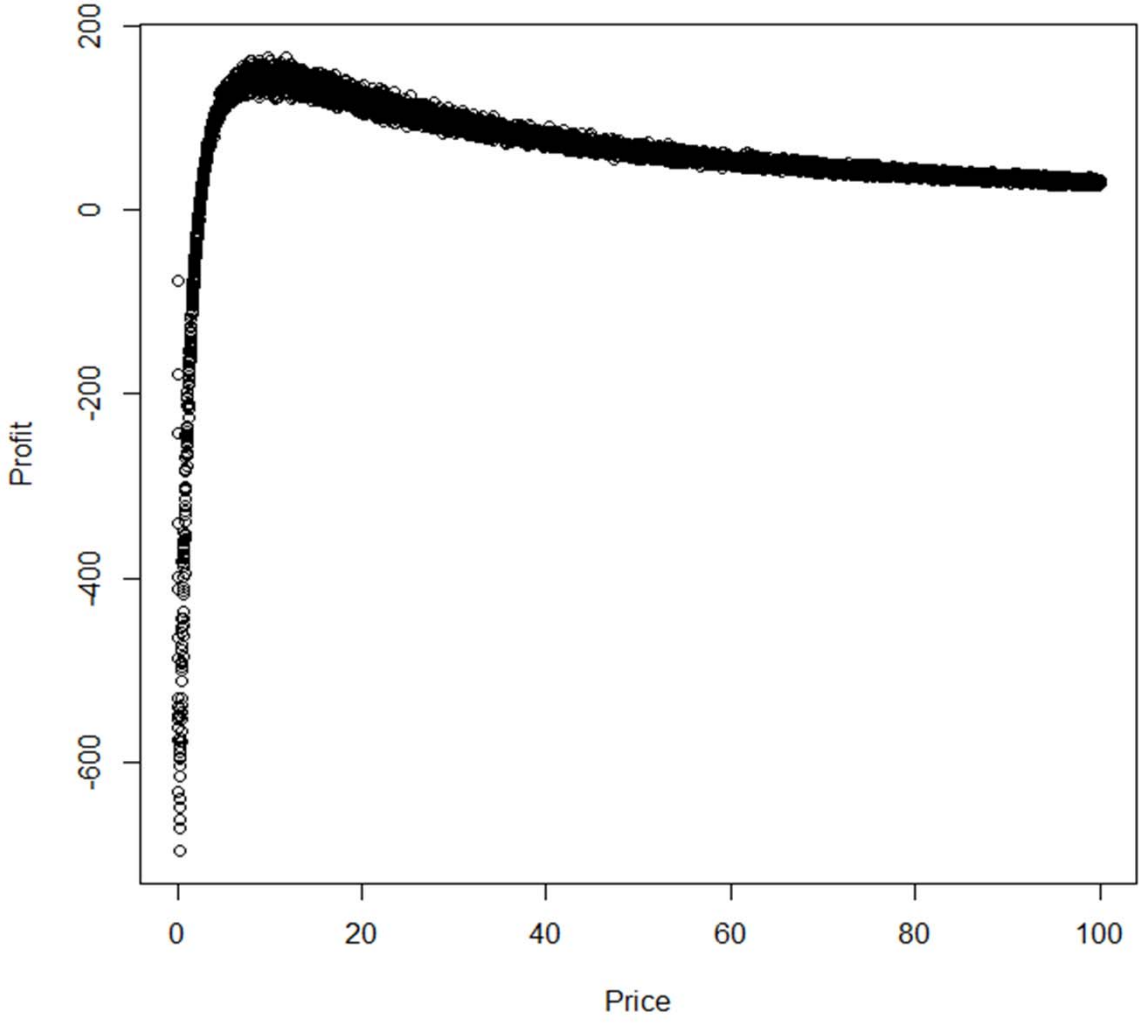
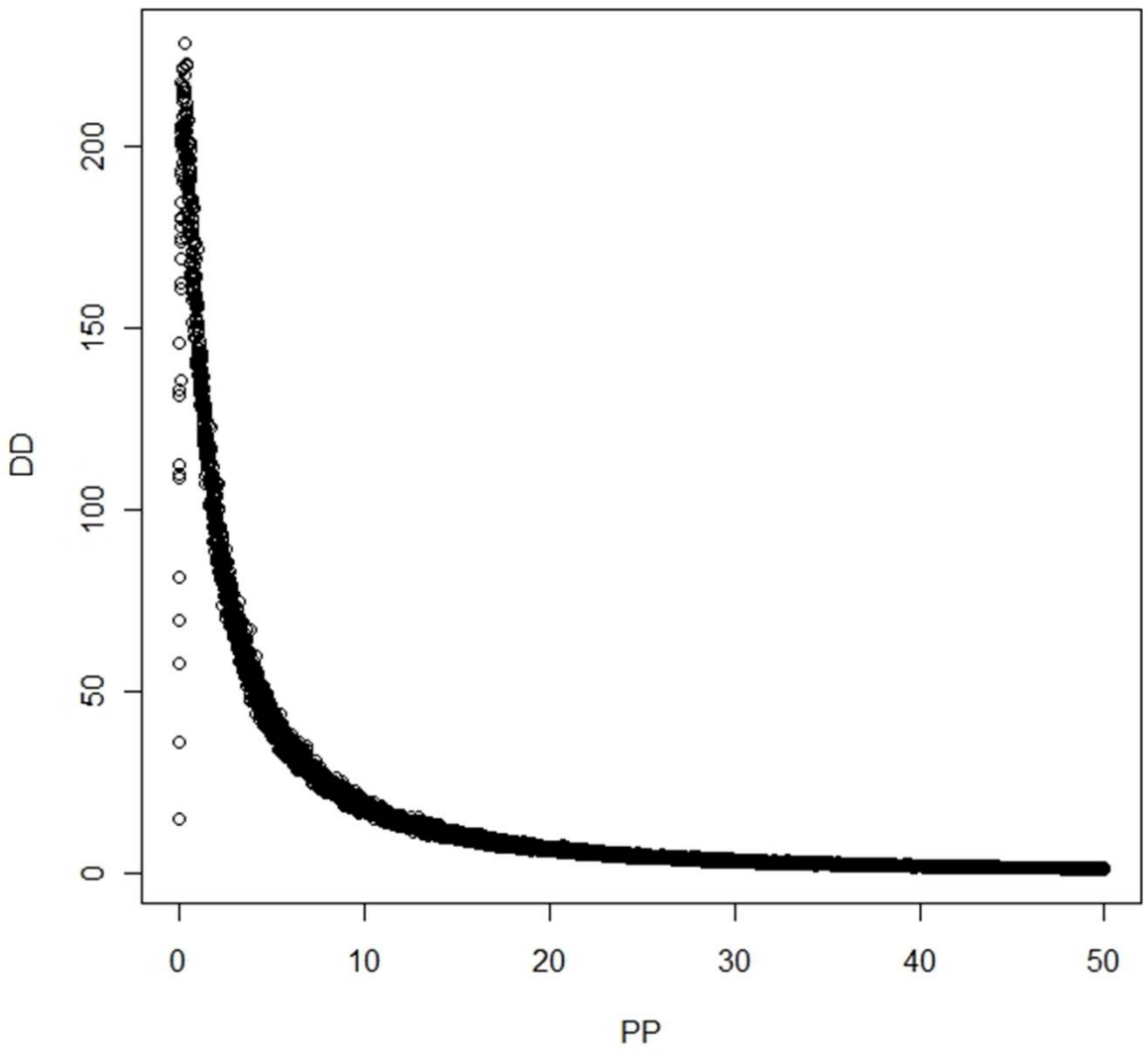
```
mykwopt1<-function (f, n=100, x0 = 15, a0 = 1, a = -1, c0 = 3, c = -1/6, B = 0.8,p=0)
{
    x <- c(x0, rep(0, n - 1))
    #pr <- c(Profit(x0), rep(0, n - 1))
    xp <- xm <- rep(x0, n)
    i=0
    for (k in 2:n) {
        p<-(p + (B^(k-2))*(f(x[k-1])))
        h <- c0 * k^c

        xp[k] <- x[k - 1] + h
        xm[k] <- x[k - 1] - h
        x[k] = x[k - 1] + a0 * k^a * (f(xp[k]) - f(xm[k]))/(2 *
            h)
        if(round(k/3)==k/3){if(k<6){

        P=c(c(x[1:k]),c(xm[1:k]),c(xp[1:k]));X=Demand(x[1:(3*k)])
        dat=data.frame(x = P, dem = X)
        lx<-log(P)
        lx2<-log(P)^2
        dem=log(Demand(P))
        f0 <- lm(dem ~ lx + lx2)
        b=f0$coef
        g<-function(x){-x*Sdemand(x,b[1],b[2],b[3])+Cost(Sdemand(x,b[1],b[2],b[3]))}
        x[k]<-nlminb(15,g,lower = .5)$par

        }}
        #pr[k] <- f(x[k])
    }
      #data.frame(x = x, xm = xm, xp = xp)
      p
}
```

# Appendix 2



# Appendix 3